

FRANÇESKA XHAKAJ

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APPOINTMENTS

2021 - present **Assistant Teaching Professor (re-appointed in 2023)**
Computer Science Department (CSD)
School of Computer Science (SCS), Carnegie Mellon University

EDUCATION

2015 - 2021 **Ph.D. Human-Computer Interaction**
School of Computer Science, Carnegie Mellon University, Pittsburgh, PA
Dissertation: Investigating How To Support Teachers In Their Teaching And Help Them Improve Their Practices Through Data And Technology.
Advisor: Amy Ogan
Thesis Committee: John Zimmerman, Geoff Kaufman, Marsha Lovett

2015 - 2017 **M.S. Human-Computer Interaction**
School of Computer Science, Carnegie Mellon University, Pittsburgh, PA

2015 - 2021 **PIER Fellowship Associate**
Program for Interdisciplinary Education Research funded by the Institute of Education Sciences, U.S. Department of Education

2011 - 2015 **B.S. Computer Science**
GPA: 3.92/4.00
Lafayette College, Easton, PA
Minor in Mathematics
Summa Cum Laude, Honors in Computer Science
Thesis: Intelligent tutors and granularity: A new approach to Red Black Trees.
Advisor: Chun Wai Liew

TEACHING EXPERIENCE

Carnegie Mellon University, Pittsburgh, PA USA

Advisor Academic advisor to 60+ CS majors, Class of 2027 [F24 - present]

Instructor **Principles of Computing (15-110)**, [S26, F25, S25, S24, F22, S22, F21]
Co-instructor (with Dr. Kelly Rivers). Across multiple offerings, taught sections ranging from 60–380 students and helped coordinate teams of 20–30 TAs.

Instructor **Introduction to Data Structures (15-121)**, [F25, F24, F23, F22, F21]
Sole instructor. Taught sections of 50-60 students with a team of 5 TAs.

Instructor **Introduction to Data Structures and Algorithms in C++ and Python**, [S25]
Designed, developed, taught a master's-level course for Information Networking Institute students.

Instructor **Introductory Programming for MHCI**, [Summer 25', 24', 23', 22']
Designed, developed, taught a master's-level course in JavaScript, HTML, and CSS for HCI students. Typical enrollment: ~30 students with 1–2 TAs each summer.

Instructor **CS Pedagogy (15-890)**, [F24, F23, S22, S20]
Co-taught seminar on computer science teaching and instructional design (with Dr. Michael Hilton/Dr. Charlie Garrod). Typical enrollment: 5-10 students per semester.

- Instructor **BHCI Capstone, Undergraduate Project in HCI (05-571), [F23-S24, F22-S23]**
Led the senior capstone for HCI majors; managed industry/community client projects. Oversaw ~60 students, 11–14 client projects, and a teaching team of 3–5 adjunct instructors and 1–2 TAs .
- Instructor **Collaborative Research (99-520): ELAIDA: Experiential Learning using AI and Data, Summer 2023**
Co-instructor (with Ganesh Mani). Mentored 5 interdisciplinary teams on AI and data projects.
- Instructor **The Role of Technology in Learning in the 21st Century (05-438/05-838), [S23]**
Taught cross-listed undergraduate/graduate course on learning and technology. (~30 students, 1 TA)
- Instructor **Principles of Computing (15-110), [Summer 2019, 2020]**
Co-instructor (with Norman Bier).
- Head TA **User-Centered Research & Evaluation (UCRE), F18, SCS, CMU**
- TA **Programming Usable Interfaces (PUI), S18, SCS, CMU**
- TA **Algorithms & Data Structures (CS150), 2013 – 2014, CS Department, Lafayette College**

INSTRUCTIONAL LEADERSHIP

Carnegie Mellon University, Pittsburgh, PA USA

- Project Supervisor **15-110 Practice Exercise Repository, S26, S25**
Supervised TA teams of 5 students in designing and building a centralized repository of supplementary practice exercises for 15-110.
- Mentor **15-121 Course Development, Summer 2024**
Mentored two students on course redevelopment, including website design and improvements to coding and pedagogical materials.
- Mentor **15-121 Course Development, F23, S24**
Mentored a master's student on the project: *Embodied cognition and tangible learning: Creating Physical Prototypes of Data Structures & Memory Concepts*.
- Faculty Mentor **BHCI Capstone Adjunct Instructors, S24, S23**
Mentored 3–5 adjunct instructors in their first teaching assignments in the BHCI Capstone course.
- Mentor **CS Pedagogy (15-890), F23, S24**
Co-mentored (with Dr. Michael Hilton) a PhD student in developing a publicly facing course website and instructional materials.

INDEPENDENT STUDIES

Carnegie Mellon University, Pittsburgh, PA USA

- Instructor **Independent Study (15-591), S26, F25, S25**
Mentoring a student in collaboration with Dr. Tom Werner (Department of Philosophy) to design and develop an app and curated sound library to support linguistics research.
- Instructor **Independent Study (15-591), S25**
Supervised a student developing an AI-assisted system to help venture capital firms more efficiently evaluate startups for investment decisions.
- Instructor **Independent Study (05-589), S24**
Mentored an undergraduate student on the project: *Exploring Gamification and Development for Confetti Social Media Platform*
- Instructor **Independent Study (05-589), Summer 2023**
Mentored a student on the project: *Melanoma Clinical Trials Connection*.

RESEARCH MENTORING

Throughout my Ph.D., I mentored exceptional undergraduate and master's students from CMU and other schools, with interdisciplinary majors (CS, HCI, design, business, statistics), many from groups underrepresented in STEM.

Undergraduate Independent Study & Research Assistants, 2018–2020

Supervised independent studies and led teams of undergraduate and master's research assistants on the ClassInSight project (30+ students overall).

Research Experience for Undergraduates (REU), Summers 2016, 2019, 2020

Mentored 12 undergraduates across projects on intelligent tutoring systems and ClassInSight.

LearnLab Summer School at CMU, ITS Track, Summer 2015 - 2018

Supervised small group projects in developing Intelligent Tutoring Systems for various domains.

PROFESSIONAL DEVELOPMENT

2018 - 2021 **Future Faculty Program**, Eberly Center Teaching Excellence & Education Innovation, CMU

2015 - 2021 **Program for Interdisciplinary Education Research (PIER)**, CMU

Fall 2018 **Computer Science Pedagogy (15-539/15-890)**, CMU

GUEST LECTURES

Fall 2019 **Graduate research seminar on Second Language Acquisition**

Department of Modern Languages, Carnegie Mellon University

Fall 2019, **Programming Usable Interfaces (PUI)**

Spring 2018 *Human-Computer Interaction Institute, Carnegie Mellon University*

RESEARCH EXPERIENCE

2019 - 2021 *Project: ClassInSight: Practical Classroom Sensing at Scale for Instructor Professional Development with Dr. Amy Ogan, Dr. Chris Harrison, Dr. Yuvraj Agarwal, Carnegie Mellon University*

2018 *Project: Improving Introductory Computer Programming Instruction by Supporting Conceptual Learning with an Intelligent Tutoring System with Dr. Vincent Aleven, Carnegie Mellon University*

2015 - 2018 *Project: Helping Teachers Help Students: Teacher's Use of Intelligent Tutoring Systems Analytics to Improve Learning with Dr. Vincent Aleven and Dr. Bruce M. McLaren, Carnegie Mellon University*

Summer 2015 *Project: Integrating Errors into the Lynnette Cognitive Tutor, an ITS for Linear Equation Solving with Dr. Vincent Aleven, Carnegie Mellon University*

Summer - Fall 2014 *Project: Integrating Intelligent Tutoring Systems in MOOCs (Massive Open Online Courses) Undergraduate Student Researcher with Dr. Vincent Aleven, Carnegie Mellon University*

Spring 2014 *Project: Creating tutors to support geology field projects EXCEL Student Researcher with Dr. Chun Wai Liew, Lafayette College*

January 2013 *Project: Computational Modeling of Fish Evolution EXCEL Student Researcher with Dr. Chun Wai Liew, Lafayette College*

Summer 2013 *Project: Studying the dynamic behavior of JavaScript objects Undergraduate Student Researcher with Dr. Barbara G. Ryder, PROLANGS@VT, Virginia Tech*

Summer 2013 *Project: Using HMM and the Viterbi Algorithm on the Iterated Diner's Dilemma game EXCEL Student Researcher with Dr. Chun Wai Liew, Lafayette College*

CONFERENCE PUBLICATIONS

- C1. **Xhakaj, F.**, Ogan, A., et al. (2021) Investigating teacher data needs in terms of teacher immediacy and nonverbal behaviors. In *Proceedings of the 15th ISLS 2021 conference*.
- C2. Ahuja, K., Shah, D., Paredy, S., **Xhakaj, F.**, Ogan, A., Agarwal, Y., Harrison, Ch. (2021) Classroom Digital Twins with Instrumentation-Free 6DOF Gaze Tracking. In *CHI 2021 conference*.
- C3. Ahuja, K., Kim, D., **Xhakaj, F.**, Varga, V., Xie A., Zhang, S., Townsend, J. E., Harrison, Ch., Ogan, A., & Agarwal, Y. (2019). EduSense: Practical Classroom Sensing at Scale. *Proc. ACM Interact. Ubiqu. Technol.*
- C4. **Xhakaj, F.**, & Aleven, V. (2018). Towards Improving Introductory Computer Programming with an ITS for Conceptual Learning. In *International Conference on AI in Education, AIED 2018*, pp. 535-538.
- C5. Bodily, R., Kay, J., Aleven, V., Davis, D., Jivet, I., **Xhakaj, F.** & Verbert, K. (2018) Open learner models and learning analytics dashboards: A systematic review. In *Proceedings of the International Conference on Learning Analytics and Knowledge, LAK 2018*, pp. 41-50.
- C6. **Xhakaj, F.**, Aleven, V., McLaren, B.M. (2017). Effects of a Teacher Dashboard for an Intelligent Tutoring System on Teacher Knowledge, Lesson Planning, Lessons and Student Learning. In *Proceedings of the European Conference on Technology Enhanced Learning, EC-TEL 2017*, pp. 315-329.
- C7. **Xhakaj, F.**, Aleven, V., McLaren, B.M. (2017). Effects of a dashboard for an intelligent tutoring system on teacher knowledge, lesson plans and class sessions. In *Proceedings of the International Conference on Artificial Intelligence in Education, EC-TEL 2017*, pp. 582-585.
- C8. **Xhakaj, F.**, Aleven, V., McLaren, B.M. (2016). How teachers use data to help students learn: Contextual Inquiry for the design of a dashboard. In *Proceedings of the European Conference on Technology Enhanced Learning, EC-TEL 2016*, pp. 340-354.
- C9. Aleven, V., **Xhakaj, F.**, Holstein, K., & McLaren, B. M. (2016). Developing a teacher dashboard for use with ITSs. In *Proceedings of the International Workshop on Teaching Analytics, EC-TEL 2016*.
- C10. Holstein, K., **Xhakaj, F.**, Aleven, V., & McLaren, B. M. (2016). Luna: A dashboard for teachers using intelligent tutoring systems. In *Proceedings of the International Workshop on Teaching Analytics, EC-TEL 2016*.
- C11. Aleven, V., Sewall, J., Popescu, O., **Xhakaj, F.**, Chand, D., Baker, R. S., & Gasevic, D. (2015). The beginning of a beautiful friendship? Intelligent tutoring systems and MOOCs. In *Proceedings of the International Conference on Artificial Intelligence in Education, AIED 2015*, pp. 525–528.
- C12. Liew, C. W., & **Xhakaj, F.** (2015). Teaching a complex process: Insertion in Red Black Trees. In *Proceedings of the International Conference on Artificial Intelligence in Education, AIED 2015*, pp. 698–701.
- C13. **Xhakaj, F.**, & Liew, C. W. (2015). A new approach to teaching Red Black Trees. In *Proceedings of the Conference on Innovation and Technology in Computer Science Education, ITICSE 2015*, pp. 278–283.

JOURNAL PUBLICATIONS

- J1. Wei, Sh., **Xhakaj, F.**, & Ryder, B.G. (2015) Empirical Study of the Dynamic Behavior of JavaScript Objects. *Journal of Software: Practice and Experience*, 46, 7, 867–889.

THESIS

- T1. **Xhakaj, F.** (2021). Investigating how to support teachers in their teaching and help them improve their practices through data and technology (Doctoral dissertation). Carnegie Mellon University.
- T2. **Xhakaj, F.** (2015). Intelligent tutors and granularity: A new approach to Red Black Trees. Unpublished senior thesis, Department of Computer Science, Lafayette College, Easton, Pennsylvania. USA.

RESEARCH TALKS

- 2018 PAWS Lab, Intelligent Systems Program: University of Pittsburgh, *Towards Improving Introductory Computer Programming with an ITS for Conceptual Learning.*
- 2018 PIER Seminar: *Supporting Conceptual Learning with an ITS for Improving Introductory Computer Programming Instruction.*
- 2018 AIED 2018 Conference: *Towards Improving Introductory Computer Programming with an ITS for Conceptual Learning.*
- 2017 PIER Seminar: *Dashboard, Dashboard On the Wall, What About My Students Don't I Know?*
- 2017 EC-TEL 2017 Conference: *Effects of a Teacher Dashboard for an Intelligent Tutoring System on Teacher Knowledge, Lesson Planning, Lessons and Student Learning.*
- 2016 EC-TEL 2016 Conference: *How Teachers Use Data to Help Students Learn: Contextual Inquiry for the Design of a Dashboard.*

EXTERNAL SERVICE

- F25, S24 Conducted outreach at Pittsburgh Obama Academy to help establish their first CS Club, provided students with intro programming experiences in the absence of a CS curriculum.
- F23 Reviewer, AERA 2024 Annual Meeting, papers in Engineering and Computer Science.
- F23 Reviewer, SIGCSE 2024 Panels track proposals.
- S23 Invited Guest Speaker, *West Windsor Plainsboro High School South (NJ)* AI Club.
- S22 Panelist, "Women & Gender Minorities in Computing: Academic Interviews," *USC*
- 2021 Reviewer, *SIGCSE 2021* Computing Education Research papers
- 2020 Panelist, Graduate School Panel for Undergraduate Women, *Georgetown University*
- 2019 CHI Reviewer, Late Breaking Work papers, *CHI 2019*
- 2018 Student volunteer, *SIGCSE 2018*

INTERNAL SERVICE

- F24 - present Academic advisor to 60+ CS majors, Class of 2027
- S25 - present Faculty Advisor, CMU Taekwondo.
- 2019 - 2024 **Co-director/co-founder of the Community for Teaching and CS Education, SCS, CMU**
Created a community for faculty and students interested in teaching and CS education.
- S24 HCII Faculty Hiring Committee
- S23 **Organizer & Panelist, Teaching in Academia Panel, SCS, CMU**
Organized a panel with educators in the Pittsburgh area to answer graduate students' questions on teaching in academia, the job market, etc.
- S24, F25 Panelist, Apple Pie with Alpha Chi Omega House, Greek Quad, *CMU*
- S23 Completed Mental Health First Aid (MHFA) training and certification
- F23 GenAI Focus Group, collaborated with Eberly Center on implications of Gen AI in education

- F22 Panelist, Eberly Teaching & Learning Summit: Alumni Panel, *CMU*
- F22 Panelist, OurCS Conference Research & Grad School Panel, *SCS, CMU*
- F22 Guest Speaker, Job Market Mentoring Group, *HCII, CMU*
- S22 **Organizer and panelist of the first *Teaching in Academia* Panel in SCS, CMU**
Organized the very first panel with teaching faculty across SCS to answer graduate students' questions on teaching in academia, the job market, etc.
- F21 Panelist representing Teaching Track, Academic Job Market Panel, *SCS, CMU*
- 2020 SCS Student Teaching Award Committee, *CMU*
- 2017, 2019 Graduate school panel moderator and conference organizer in the OurCS conference
- 2017 - 2019 **Women in SCS**, leader and organizing committee, *CMU*
- 2019 - 2020 Student representative, HCII **Faculty Hiring Committee** and HCII **Curriculum Committee**
- 2016 - 2019 Organized multiple PhD student events (Open House, Social Events), *HCII, CMU*
- 2012 - 2015 Founder and President, Women in Computing, *Lafayette College*
- 2014 - 2015 President of Upsilon Pi Epsilon International Honor Society for the Computing and Information Disciplines, *Lafayette College*

AWARDS AND HONORS

- 2019 [Alan J. Perlis Graduate Student Teaching Award](#), *Carnegie Mellon University*
Recognizes exceptional dedication and excellence in teaching within the School of Computer Science.
- 2019 Graduate Student Assembly Departmental Appreciation Award, *Carnegie Mellon University*
Honors outstanding contributions to the department community.
- 2015 Summa Cum Laude, Honors in Computer Science, *Lafayette College*
- 2014 UPE Special Recognition Award, *Upsilon Pi Epsilon International Honor Society for the Computing and Information Disciplines*
- 2014 James P. Schwar Prize, *Lafayette College*
- 2011 - 2015 Walter Oechsle Scholarship, *Lafayette College*
- 2012, 2013 Grace Hopper Celebration of Women in Computing Scholarship, *Grace Hopper Conference*